Shiva Sai Pagilla

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.97
Narayana Junior College Chaitanyapuri, Saroornagar	Class XII (TSBIE)	2023	97.90%
Kendriya Vidyalaya No. 1 Uppal	Class X (CBSE)	2021	91.20%

WORK EXPERIENCE

Forward Deployed Engineer Intern

Nurix AI

- Built and deployed **a production voice agent** for **outbound lead qualification** for Aditya Birla Capital Limited (ABCL), handling **~1 lakh leads** at **100-call concurrency**; tuned combinations of **LLM, STT, TTS, and voice configuration** to optimize **response accuracy and call latency**.
- Designed a **post-call analysis process** to extract key metrics from call transcripts, including connectivity status and **L1/L2 dispositions**, and built **SQL-based Metabase dashboards** tracking conversion rate, drop-off points, and hourly/weekly calling funnel trends, used by the client to guide agent performance improvements.
- Integrated **MCP into the voice agent** to trigger dynamic API calls during on-call, and post-call stages, extending the agent from conversation into **action-taking**.
- Built an **event-driven serverless pipeline** using **AWS Lambda** and **SNS** processing real-time post-call data, updating connected-call statuses for **30%** of dialed leads to the client's **Salesforce CRM via REST APIs**; also automated a **cron-scheduled Lambda pipeline** for batch data transfer to the client's GCP environment, replacing manual delivery.
- Developed an **omnichannel conversational AI demo** for **Aditya Birla Health Insurance**, integrating **website chatbot, WhatsApp, and voice agent** into a single flow to re-engage dropped-off leads and **demonstrate cross-platform capabilities**.

PROJECTS

AI Quant Agent | Github

- Built a multi-agent financial analysis platform with **FastAPI, LangGraph, and LangChain**, aggregating real-time **yfinance** market data to orchestrate LLM agents for structured Buy/Sell recommendations.
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline that embeds and retrieves relevant news articles from a **Qdrant** vector database to ground LLM analysis in up-to-date context.
- Integrated an in-memory **TTL caching layer** to reduce LLM requests and drastically improve API response latency.

MiniChain | Github

- Developed a modular **blockchain system in Python** with transaction, block, mining, and consensus layers.
- Built a **TCP/asyncio peer-to-peer network** for block propagation, mempool management, chain synchronization, and fork resolution.

Medicine Recommender System Github

- Designed a medicine recommender system that suggests alternative medicines based on similarity in composition and therapeutic use.
- Utilised data processing techniques, Word2Vec embeddings, and a weighted similarity algorithm to capture semantic relations and improve matching accuracy.
- Generated a ranked list of 10 closest medicine alternatives, supporting healthcare providers and patients in identifying cost-effective and accessible options.

SKILLS

Languages: C++, Python, SQL, Javascript

ML/Data Science: Numpy, Pandas, Sklearn, Matplotlib, Tensorflow, Keras

GenAI: LangChain, LangGraph, Hugging Face, Multi-Agent Systems, RAG, LLM APIs, Prompt Engineering

Other Tools: Git, Github, FastAPI, PostgreSQL, REST APIs, some basic AWS services.

COURSES AND CERTIFICATIONS

Mathematics: Elementary Linear Algebra, Calculas, Differential Equations, Probability

Computer Science: Machine Learning, Data Structures & Algorithms, Artificial Intelligence. Introduction to Programming.

Certifications: Summer Analytics - Data Science & Machine Learning, IIT Guwahati **| Advanced Learning Algorithms,** Stanford University

POSITIONS OF RESPONSIBILITY

Mentor (prev. Overall Head), Blockspace - The Blockchain & Crypto Club, IIT Hyderabad

Sponsorship Coordinator, Milan General Championship, IIT Hyderabad

Events & PRO Coordinator, Tinkerers' Lab, IIT Hyderabad

Operations Manager, Entrepreneurship Cell, IIT Hyderabad

EXTRACURRICULAR

- Secured 3rd place in a **Web Development Hackathon** in institute's **Inter-Hostel Competition.**
- Active participant in Football, Cricket, and Volleyball
- Volunteered in **National Service Scheme(NSS)** Events